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Influence of Interlimb Lean Muscle Mass Asymmetry on Countermovement Jump Neuromuscular Performance Qualities Among American Football Players

Johnathan Polletta,[1] Kevin Leung,[1] David Diaz,[1] Nicole Branum,[2] and Monique Mokha[1]

[1]Department of Health and Human Performance, College of Health Care Sciences, Nova Southeastern University, Ft. Lauderdale, Florida; and [2]Nutrition Program, College of Osteopathic Medicine, Nova Southeastern University, Ft. Lauderdale, Florida

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Abstract

Polletta, J, Leung, K, Diaz, D, Branum, N, and Mokha, M. Influence of interlimb lean muscle mass asymmetry on countermovement jump neuromuscular performance qualities among American football players. *J Strength Cond Res* XX(X): 000–000, 2024—Body composition is a determinant of athletic performance. The purpose of this study was to determine the influence of lower limb lean muscle mass (LMM) asymmetry on countermovement jump (CMJ) performance and kinetic asymmetry during both eccentric and concentric CMJ phases. Seventy-four American football players (age, 23.0 ± 0.9 years; height, 1.86 ± 0.07 m; mass, 104.9 ± 21.6 kg; lean body mass, 88.7 ± 12.5 kg) training for the 2024 National Football League draft underwent body composition assessment using an InBody 270 bioelectrical impedance analyzer that yielded right and left leg LMM in kg. Subjects then performed 3 CMJs on dual uniaxial force plates that allowed for the calculation of jump height (JH), reactive strength index modified (RSImod), and interlimb percentage asymmetry of peak force and impulse for eccentric and concentric phases. A paired *t*-test and Pearson Product-moment Correlation Coefficients calculated the difference between left and right LMM and the associations between the symmetry variables, respectively, $p \leq 0.05$. Right and left leg LMM were significantly different ($t(73) = 3.772$, $p \leq 0.001$). However, no significant relationships were found between lower limb LMM asymmetry (0.67 ± 0.67%) and any asymmetry in CMJ eccentric peak force (4.77 ± 3.75%), $r(72) = 0.001$, $p = 0.954$; eccentric deceleration impulse (5.23 ± 4.36%), $r(72) = 0.158$, $p = 0.180$; concentric peak force (2.99 ± 2.54%), $r(72) = -0.70$, $p = 0.553$; concentric impulse (3.06 ± 2.97%), $r(72) = 111$, $p = 0.347$; JH (41.8 ± 7.3 cm), $r(72) = -0.205$, $p = 0.08$; or RSImod (0.53 ± 0.14 $m \cdot s^{-1}$), $r(72) = -0.151$, $p = 0.199$. The magnitude of LMM asymmetry did not influence CMJ performance or asymmetry magnitude. Reducing CMJ kinetic asymmetry may require targeted programming, especially for the eccentric loading phase.

Key Words: body composition, CMJ kinetics, impulse, eccentric phase

Introduction

American football is a collision-based sport that requires a unique combination of size, strength, speed, and power (40). Players compete in different positions that have specific technical, tactical, and physical activity demands (22,24,26,30) and are recognized for their large anthropometric characteristics (2,40). Various anthropometric indices such as muscle volume, total body mass (TBM), and lean muscle mass (LMM) have been associated with joint torque (18) and tests of football-related performance (e.g., 1 repetition maximum bench press, 20- and 40-yard run times, vertical jump height [JH]) (30), discriminate between player positions (17), affect career earnings and longevity (35), and undergo increases through training (2,26). Thus, the optimization of body composition and size in American football players is a well-known priority. This is especially apparent in the lineman, linebacker, and tight end positions where the accrual of LMM is emphasized. Body fat percentage (BF%) is also an important factor affecting performance where higher body fat negatively influences vertical JH (21,30), power clean load (30), and 20- and 40-yard run times (30) in American football players. Interestingly, Haizlip et al. (21) showed that vertical JH and reactive strength were highly negatively associated with LMM and TBM as well as fat mass in National Football League (NFL) draft preparation camp subjects. They concluded that heaviness was inversely related to explosiveness in the vertical jump and cautioned to consider performance effects when emphasizing mass accrual. Annual NFL combine results also show that heavier players like those in the Big positions (e.g., offensive and defensive linemen) do not jump as high as lighter players in the Skill positions (e.g., wide receivers, cornerbacks) (34). In men and women athletes of different sports (i.e., soccer, basketball), healthy adults and youth, and recreationally active young adults, LMM has been directly associated with strength, power, and/or jump performance (6,7,25,32,39). Given the relationship between player anthropometrics and performance, teams allocate considerable resources to strength and conditioning programs to improve and monitor TBM, LMM, and BF%. To that end, interlimb symmetry has become a metric of interest with the assumption that a certain magnitude of asymmetry is expected depending upon limb dominance, sport laterality, and/or medical history (3,7,10,16,32,30). Assessing interlimb asymmetries is common practice in clinical settings to determine return to play and/or identify injury risk (5,31–34) with less than 10% and greater than 15%, respectively, suggested as thresholds (8). While assessment

Address correspondence to Monique Mokha, gm588@nova.edu.

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of interlimb asymmetries is becoming common practice in strength and conditioning, the influence on athletic performance is less known. Yoshioka et al. (45) examined the effect of muscle strength asymmetry on the maximal JH using computer simulation. Jump heights from the model with a 10% strength asymmetry were not significantly different (<1%) than the model without asymmetry. Conversely, Bell et al. (7) found that thigh and shank LMM asymmetry explained 20% of the variance in countermovement jump (CMJ) force asymmetry, while LMM asymmetry of the pelvis, thigh, and shank explained 25% of the variance in CMJ power asymmetry in a group of National Collegiate Athletic Association Division 1 athletes that included a subset of American football players. Furthermore, subjects with a 10% or greater asymmetry in CMJ power had reduced JH of up to 8.9 cm (effect size of $d > 0.8$), illustrating an influence of symmetrical power application for CMJ performance. In a 2018 systematic review of asymmetry and sports performance (8), the authors concluded that while the prevalence of asymmetries in morphology, strength, and performance measures (e.g., jumping, dynamic balance, sprinting, kicking) in athletes is high, outcomes are inconsistent. Regarding morphology and CMJ performance, inconsistencies may be due to CMJ type (e.g., single or double leg, use of an arm swing) and variable of interest (e.g., JH, peak power, reactive strength) and/or morphology measurement type (e.g., bioelectrical impedance, dual-energy x-ray absorptiometry, girth) and variable of interest (e.g., fat mass, LMM, BF%). The equation used to calculate symmetry (or asymmetry) also contributes to inconsistencies (9).

American football involves activities such as accelerations, decelerations, changes of direction, and jumping that are influenced by the interaction of morphological and neuromuscular factors (4). Power is defined as a rate of performing mechanical work or a product of force acting upon an object and the object's velocity. The CMJ is one of the most implemented testing modalities for assessing lower body power and other mechanical capacities (12–14,27,28,40). The CMJ provides feedback to strength and conditioning coaches about sports performance (28), interlimb asymmetries (7,8), neuromuscular fatigue (11), and the effectiveness of training programs (41). Power output during a CMJ and the JH achieved are relevant indicators of lower limb performance in athletes (17). Hence, the vertical jump, a CMJ performed with an arm swing, is one of the 7 physical performance tests at the NFL scouting combine. Ideally, throughout a CMJ, the athlete will produce relatively equal force and power bilaterally to enhance performance and reduce the potential for injury. To date, researchers have not discriminated interlimb asymmetry by CMJ phase relative to morphology. The braking or loading phase of the CMJ is where the athlete downwardly decelerates their center of mass in active lengthening of most of the lower limb muscles (29). Eccentric strength is a critical aspect of the ability to decelerate, stop, change direction, and absorb the high loads associated with football's explosive movements. Specific to the CMJ, eccentric variables explain 16% of CMJ variance versus 11% from concentric metrics (1). Given the difference between the braking and propulsion phases in parameters such as predominant muscle contraction type (eccentric vs. concentric), eccentric:concentric time, and rate of force development, examining results by phases might better inform training for performance and injury prevention by identifying deficiencies. Furthermore, as high-level players rely heavily on explosive power to excel in certain field requirements where the slightest physical discrepancy among similar caliper opponents can significantly affect play, scrutiny of interlimb asymmetry appears crucial. Therefore, the purpose of this study was to determine the relationship between leg LMM and CMJ asymmetries in a group of American football players preparing for the NFL draft. Results may influence strength and conditioning practices by illuminating the influence of body composition asymmetries on performance asymmetries in highly trained athletes.

Methods

Experimental Approach to the Problem

This was a cross-sectional correlational study designed to examine body composition and functional characteristics of collegiate American football players training for the NFL draft in January 2024, of which 35% had combined invitations. Data were collected during the first week of a 6-week training camp. Body composition, specifically lower limb LMM, was collected using an InBody 270 (InBody USA, Cerritos, CA), a multifrequency bioelectrical impedance analyzer (BIA). A dual x-ray absorptiometry (DXA) device was available to the researchers. However, due to manufacturer weight (125 kg) limitations, it was not practical with this sample as 15 subjects weighed over 125 kg. Interdevice associations between DXA (gold standard) and InBody 270 variables have been reported as 0.96–0.99 with high relative reliability (intraclass correlation coefficient [ICC] = 0.98–1.00) (27). Using 47 exercise-trained adults, Czartoryski et al. (15) reported underestimations of BF% and fat mass and overestimations of fat-free mass by the InBody 270 compared with DXA. However, none of these estimations were significantly different leading the researchers to deem both devices acceptable for body composition analysis. Jump height, reactive strength, peak force, and impulse during both the eccentric and concentric phases of the CMJ were collected using dual uniaxial force plates (VALD, Brisbane, Australia). The plates have been reported to show good to excellent validity (<10% difference) compared with laboratory force plates and good to excellent reliability

Table 1
Description of countermovement jump variables.

Variable	Units	Definition
Jump height	cm	Maximal displacement of the subject's center of mass (CoM); (take-off velocity)^2/2(9.81m·s^{-2})
Reactive strength index, modified	m·s^{-1}	Ability to change quickly from eccentric to concentric contraction; jump height/contraction time
Normalized peak power	W·kg^{-1}	Highest power output during the jump relative to a subject's body mass
Concentric impulse asymmetry	%	Absolute difference in right and left limbs in total work from the point of the lowest portion of the CoM to the point of take-off or the propulsion phase
Eccentric impulse asymmetry	%	Absolute difference in right and left limbs in total work from the point the movement starts to the lowest point of the CoM depth

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(ICCs >0.75) (12). Differences in lower limb LMM were found to be significantly different providing impetus to conduct the examination.

Subjects

Seventy-four male collegiate American football players ages 21–25 years with mean ± SD height of 1.86 ± 0.07 m; mass of 104.9 ± 21.6 kg; and LMM of 88.7 ± 12.5 kg participated in this study. To be included in the study, the subjects must have attended a specialized camp preparing for the NFL draft, be at least 18 years old, and be able to complete a CMJ without any medical restrictions. All subjects had recently completed their collegiate competitive football season, were active players training 5–6x per week, and were experienced in performing maximal effort CMJs. The data set included all male collegiate football players from various universities and in-game positions. Subjects arrived at the Sports Performance and Gait Laboratory at Nova Southeastern University in groups of 3 at a designated time. All data were collected over 2 testing days. Players were advised of the risks and benefits of participation, and written informed consent was obtained before participation. The university's Institutional Review Board approved this study (#2018-684).

Procedures

Total body mass (kg), LMM (kg), right and left limb LMM, and BF% were measured with an InBody 270 multifrequency (BIA) device (InBody USA) that was calibrated before testing. InBody 270 is automatically calibrated upon device activation and using its level indicator to ensure that the unit is balanced. Subjects were instructed to void before testing and refrain from eating, drinking, or exercising for at least 3 hours before any testing. Procedures were in accordance with the manufacturer's guidelines where subjects stood on the platform of the device barefoot with the soles of their feet on the electrodes. Subjects then grasped the handles of the unit with the thumb and fingers maintaining direct contact with the electrodes. The elbows were fully extended, and the glenohumeral joints were abducted to approximately 30°. The duration of the test was less than 1 minute.

Subjects performed 1–2 submaximal practice CMJ using a dowel held horizontally across the posterior shoulders. Then, 2 maximal effort CMJs with the dowel were performed and recorded on dual uniaxial force plates (FD Lites; VALD) sampling at 1,000 Hz. ForceDecks software (ver. 1.8.6; VALD) was used to calculate averages of both CMJ for the variables that are described in Table 1. The force plates were calibrated before each

Table 2

Mean ± *SD*s and 95% confidence intervals for left and right limb lean muscle mass and countermovement jump values.

	Left	Right
Lean muscle mass, kg	12.59 ± 1.56 (12.23–12.95)	12.69 ± 1.59 (12.32–13.05)*
Eccentric peak force, N	1,209.9 ± 274.7 (1,146.3–1,273.6)	1,202.9 ± 267.8 (1,140.9–1,265.0)
Eccentric impulse, $N \cdot s^{-1}$	172.9 ± 45.2 (162.5–183.5)	170.7 ± 48.8 (159.4–182.0)
Concentric peak force, N	1,321.9 ± 229.2 (1,268.8–1,375.1)	1,307.4 ± 215.5 (1,257.5–1,357.4)
Concentric impulse, $N \cdot s^{-1}$	292.9 ± 62.9 (278.3–307.5)	293.6 ± 64.6 (278.0–308.6)

*indicates $p < .001$

Table 3

Mean, *SD*s, 95% confidence intervals, and Pearson correlation matrix for asymmetries in interlimb lean muscle mass and countermovement jump variables.*

	Mean	*SD*	Lean muscle mass, %	Eccentric peak force, %	Eccentric impulse, %	Concentric peak force, %	Concentric impulse, %	RSImod, $m \cdot s^{-1}$	Jump height, cm
Lean muscle mass, %	0.67 (0.5–0.8)	0.66							
Eccentric peak force, %	4.8 (3.9–5.6)	3.7	0.007 (−0.228 to 0.242)						
Eccentric impulse, %	5.3 (4.3–6.3)	4.4	0.158 (−0.074 to 0.390)	0.755† (0.601 to 0.909)					
Concentric peak force, %	2.98 (2.4–3.6)	2.5	−0.070 (−0.305 to 0.164)	0.702† (0.534 to 0.869)	0.615† (0.430 to 0.801)				
Concentric impulse, %	3.1 (2.4–3.7)	3.0	0.111 (−0.123 to 0.344)	0.694† (0.524 to 0.863)	0.667† (0.492 to 0.842)	0.696† (0.528 to 0.865)			
RSImod, $m \cdot s^{-1}$	0.53 (0.5–0.6)	0.14	−0.151 (−0.383 to 0.081)	−0.275†(−0.501 to −0.049)	−0.241† (−0.469 to −0.013)	−0.216 (−0.445 to 0.013)	−0.426† (−0.638 to −0.213)		
Jump height, cm	41.8 (40.1–43.5)	7.2	−0.205 (−0.435 to 0.025)	−0.288‡ (−0.513 to −0.063)	−0.244‡ (−0.472 to −0.016)	−0.270† (−0.497 to −0.044)	−0.415† (−0.628 to −0.201)	0.834† (0.704 to 0.964)	

*RSImod = reactive strength index modified.
†indicates $p < 0.001$
‡indicates $p < 0.05$

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trial for each subject according to manufacturer's guidelines (41) that included zeroing the plates and weighing the subject. Familiarity with force plates by the subjects was varied and was not documented for this study. ForceDecks uses a 20-N offset from the measured bodyweight, which was quantified before all jump trials to define the start of the movement. The end of the eccentric phase and the start of the concentric phase were defined as minimum absolute displacement (zero velocity), and take-off was defined as the time point at which the total vertical force fell below the threshold of 30 N below bodyweight (29). Interlimb asymmetries in eccentric and concentric peak force and impulse (IMP) were calculated using the equation {([high − low]/[high + low]) × 100}. The investigators chose this formula since the data would not be collected over extended periods potentially resulting in loss of context if different limbs produced the highest score (8).

Statistical Analyses

All statistical analyses were processed using the Statistics Package for Social Sciences (version 29; IBM Corporation, New York, NY). Descriptives (mean ± *SD*s; 95% confidence intervals) were calculated for overall and interlimb data from body composition and CMJ measurements. Pearson's correlation coefficients were calculated to determine associations between interlimb LMM and CMJ eccentric and concentric peak force and IMP asymmetries, $p < 0.05$. Correlation strength was determined according to Schober et al. (38) and is as follows: negligible: $r < 0.10$; weak: $r = 0.10$–0.39; moderate: $r = 0.40$–0.69; strong: $r = 0.70$–0.89; and very strong: $r = 0.90$–1.00. Cohen's *d* effect size was also calculated, with the magnitude of effect classified in accordance with Hopkins (23) as follows: $d \geq 0.2$ is small; $d \geq 0.6$ is moderate; $d \geq 1.2$ is large; $d \geq 2.0$ is very large, and $d \geq 4.0$ is nearly perfect.

Results

Table 2 presents the mean and *SD*s of the performance variables (JH, reactive strength index modified [RSImod], and countermovement depth). Table 3 shows the LMM and CMJ characteristics for left and right sides. Right and left leg LMM were significantly different ($t(73) = 3.772$, $p \leq 0.001$. However, no significant relationships were found between lower limb LMM asymmetry (0.67 ± 0.67%) and any asymmetry in CMJ eccentric peak force (4.77 ± 3.75%), $r(72) = 0.001$, $p = 0.954$, eccentric deceleration impulse (5.23 ± 4.36%), $r(72) = 0.158$, $p = 0.180$, concentric peak force (2.99 ± 2.54%), $r(72) = -0.70$, $p = 0.553$, concentric impulse (3.06 ± 2.97%), $r(72) = 111$, $p = 0.347$, JH (41.8 ± 7.3 cm), $r(72) = -0.205$, $p = 0.08$, or RSImod (0.53 ± 0.14 $m \cdot s^{-1}$), $r(72) = -0.151$, $p = 0.199$. See Figures 1–4.

Discussion

We sought to determine if asymmetries in LMM between the lower limbs would influence CMJ height, reactive strength, peak force output, and impulse asymmetry during both the eccentric and concentric phases of a CMJ. Our findings indicate that the asymmetries found in LMM were not directly associated with any of the CMJ asymmetries found regarding the variables in JH and RSImod. Body composition limb differences are not significantly related to CMJ limb differences, but CMJ differences are negatively associated with performance such as JH or the reactive strength index modified (RSImod). Mean JH was 41.8 cm with a *SD* of 7.3 cm. RSI-mod had a mean of 0.53 $m \cdot s^{-1}$ with a SD of 0.14 $m \cdot s^{-1}$.

The lack of association may be due to the overall low LMM asymmetries among the players. In fact, some players were symmetrical in their LMM, yet their results still indicated asymmetries, showing that although a player can be symmetrical in their LMM, some other factor is causing asymmetrical jump data and results. This is also apparent since there were also a small number of players who had asymmetries in their LMM but were evenly balanced in the CMJ variables. Data showed that while left and right lower limb LMM were significantly different, mean asymmetry values were low at < 1%.

Others have investigated the relationship between LMM imbalances and functional asymmetry. Bell et al. (7) reported that LMM asymmetry of the thigh and shank explained 20% of the variance in CMJ peak force. In Australian footballers, inaccurate kickers had greater LMM asymmetry than accurate kickers, and an asymmetry of 3% in LMM created an 8% imbalance in isometric leg strength (20). Chapelle et al. (10) found no significant relationship between LMM asymmetry and functional (e.g., hand grip, single leg CMJ, 505 change of direction) asymmetry in elite youth tennis players. The latter 2 investigations focused on athletes with sport laterality and limb dominance. This study's strength is the inclusion of asymmetry for the eccentric phase of a CMJ in addition to the commonly reported concentric phase. This led to the finding that larger asymmetries were present

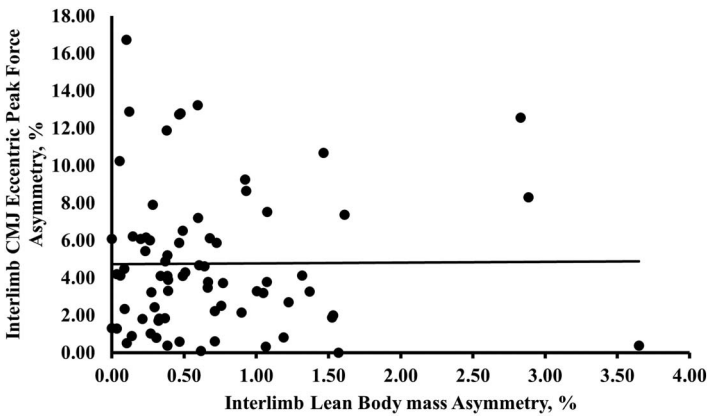

Figure 1. Association between interlimb asymmetries in lean body mass and CMJ eccentric peak force. CMJ = countermovement jump.

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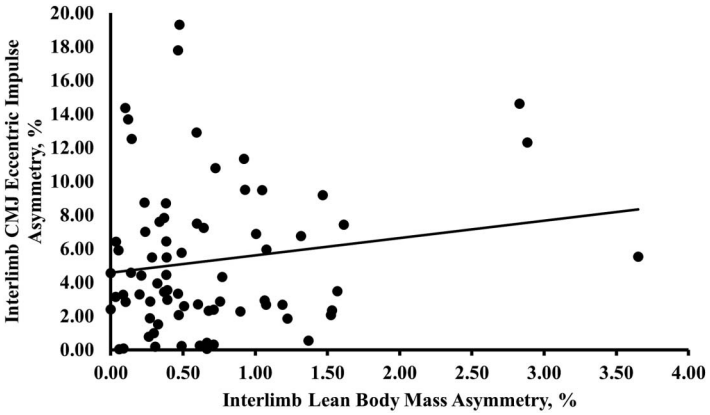

Figure 2. Association between interlimb asymmetries in lean body mass and CMJ eccentric impulse. CMJ = countermovement jump.

during the eccentric (peak force, 4.8%; impulse, 5.3%) versus concentric phase (peak force, 2.98%; impulse, 3.1%) where greater neuromuscular coordination is required to control the deceleration of the body's center of mass (CoM) (19,42,43). Neuromuscular coordination, involving muscular strength, activation timing, and synergistic function (37), may have influenced the lack of association between the body composition and CMJ asymmetries. A few studies have examined the effects of neuromuscular control on interlimb asymmetries for jumping tasks in athletes. Thus, this finding warrants further investigation.

Player position requirements may also have influenced the results of this study. Offensive and defensive players engage in different types of directional movements possibly resulting in different asymmetries (18). Players may develop performance asymmetry resulting from the position-specific movements (18). This is an important consideration as some of the asymmetries developed may have greater influence on jump performance than others.

Interlimb asymmetry in athletes is common, especially in many youth athletes (36). It is commonly known that an individual's dominant limb is stronger than their nondominant limb. Studies have also shown that interlimb differences for functional movement have been established in early childhood, and measures must be taken to ensure that the asymmetry does not increase (36). These asymmetries can lead to additional stress on the player, which may lead to a drop in performance for jumping tasks (36). Thus, limb dominance may have influenced the results of this study. The limb with greater LMM may be stronger and have led to larger force and impulse. However, studies have also shown that limb dominance has a minimal effect since the dominant limb will compensate for the strength deficit of the nondominant limb (45).

The lack of a control group or comparison group makes it difficult to isolate LMM asymmetries that are specific to American football players. Without a direct comparison or control group, there are challenges to determine whether any observed asymmetries are sport specific or a result of other factors such as natural asymmetries, training regiments, injury history, or positional differences. To better isolate the effect of LMM asymmetry on performance metrics in American football, any future studies include a control group of athletes from other sports or a population of individuals with similar activity levels, but no specialized training to help clarify if the asymmetries are sport specific.

While this study highlights the complexity of the LMM influences regarding CMJ performance metrics, future research is needed to explore the impacts of injury history, limb dominance, neuromuscular control training, and positional requirements on CMJ performance metrics in highly skilled American style football players.

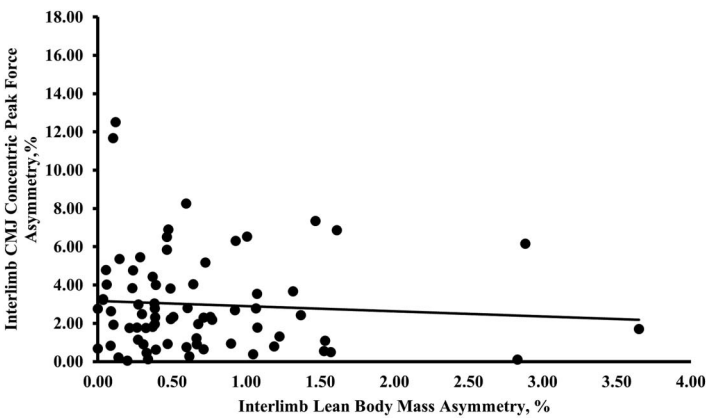

Figure 3. Association between interlimb asymmetries in lean body mass and CMJ concentric peak force. CMJ = countermovement jump.

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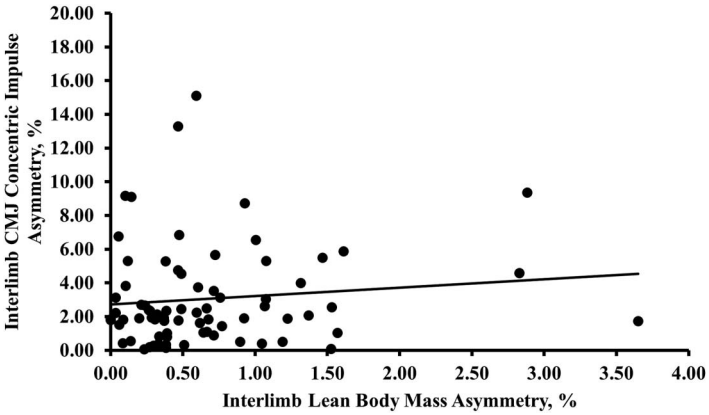

Figure 4. Association between interlimb asymmetries in lean body mass and CMJ concentric impulse. CMJ = countermovement jump.

Practical Applications

Optimizing body composition for player health and performance is commonplace for strength and conditioning specialists. This includes developing, maintaining, or restoring body composition symmetry. In addition, results of this study indicate that interlimb differences in LMM are not associated with the differences in peak force and impulse output during a CMJ. Thus, while body composition management should continue, it cannot be assumed that LMM symmetry translates to equal performance by both limbs during a bilateral task. As results of this study show, even highly skilled athletes familiar with a CMJ possess asymmetries that may affect performance. Higher jumping is predicated on the equal application of force in the eccentric and concentric phases. Those athletes with an equal application of force jumped higher than those who did not have an equal application of force in the jump during performance. This may be related to neuromuscular activation and coordination instead of LMM symmetry alone. This was especially evident for the eccentric loading phase of the CMJ where the athlete decelerates their COM. Practitioners may need to train symmetrical force and power production, especially in the eccentric phase of the CMJ. For players with low CMJ performance (height) and greater LMM asymmetry, the practitioner may implement unilateral strength exercises targeting the weaker leg to attempt and reduce the asymmetry. For players with low CMJ performance and little to no asymmetry, practitioners may select more bilateral power exercises to increase jump performance. Practitioners may also consider short-term neuromuscular training (NMT) in training programs to allow these athletes to improve jump performance metrics with a focus on the eccentric phase. Short-term NMT programs will allow the athlete to potentially improve jump landing mechanics and enhance performance in explosive jumping tasks (44). Implementation of jump training tasks paired with sprint training tasks and strength training tasks over 6 weeks in 1-hour sessions can lead to improvements for jump performance metrics (44). Other NMT techniques should be focused on the eccentric deceleration or loading phase as there is greater neuromuscular control throughout the eccentric phase compared with the concentric phase. The use of force plates allows for the monitoring of asymmetry during all phases of the CMJ and would be useful to identify deficits, individualize training, and monitor progress.

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